Environmental Chemistry

Question1

Which of the following is true about freons?
[24-Jan-2023 Shift 1]

Options:

A. These are chlorofluorocarbon compounds

B. These are chemicals causing skin cancer

C. These are radicals of chlorine and chlorine monoxide

D. All radicals are called freons

Answer: A

Solution:

Fact

Question2

Correct statement is :
[24-Jan-2023 Shift 2]

Options:

A. An average human being consumes more food than air

B. An average human being consumes nearly 15 times more air than food

C. An average human being consumes equal amount of food and air

D. An average human being consumes 100 times more air than food

Answer: B

Solution:

Theoretical.

Question3

Correct statement about smog is
[29-Jan-2023 Shift 1]

Options:

A. NO_2 is present in classical smog

B. Both NO_2 and SO_2 are present in classical smog

C. Photochemical smog has high concentration of oxidizing agents

D. Classical smog also has high concentration of oxidizing agents

Answer: C

Solution:

Photochemical smog has high concentration of oxidising agents
NO_2 is produced from NO and O_3 in the presence of sunlight
Classical smog contain smoke, fog and SO_2 and it is known as reducing smog, as chemically it is reducing mixture

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Question4

The concentration of dissolved Oxygen in water for growth of fish should be more than xX ppm and Biochemical Oxygen Demand in clean water should be less than xY ppm. X and Y in ppm are, respectively.
[29-Jan-2023 Shift 2]

Options:

A. X Y
6 5

B. X Y
4 8

C. X Y
4 15

D. X Y
6 12

Answer: A

Solution:

Solution:
The growth of fish gets inhibited if the concentration of dissolved Oxygen in water is less than $6\,\text{ppm}$ and Biochemical Oxygen demand in clean water should be less than $5\,\text{ppm}$.

Question5

Formation of photochemical smog involves the following reaction in which A, B and C are respectively.

(i) $NO_2 \xrightarrow{hu} A + B$

(ii) $B + O_2 \rightarrow C$

(iii) $A + C \rightarrow NO_2 + O_2$

Choose the correct answer from the options given below:
[30-Jan-2023 Shift 1]

Options:

A. O, NO&NO_3^-

B. O, N_2O& NO

C. N, O_2&O_3

D. NO, O&O_3

Answer: D

Solution:

$$NO_{2g} \xrightarrow{hv} \underset{(A)}{NO_g} + \underset{(B)}{O_g}$$

$$\underset{(g)}{O_g} + O_{2g} \rightleftharpoons \underset{(C)}{O_{3g}}$$

$$\underset{(A)}{NO_g} + O_3 \underset{(C)}{g} \rightarrow NO_{2g} + O_{2g}$$

Question6

The water quality of a pond was analysed and its BOD was found to be 4 . The pond has
[30-Jan-2023 Shift 2]

Options:

A. Highly polluted water

B. Water has high amount of fluoride compounds

C. Very clean water

D. Slightly polluted water

Answer: C

Solution:

Solution:
Clean water as BOD value of <5 while polluted water has BOD of 15 or more.

Question7

The normal rain water is slightly acidic and its pH value is 5.6 because of which one of the following ?
[31-Jan-2023 Shift 2]

Options:

A. $CO_2 + H_2O \rightarrow H_2CO_3$

B. $4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$

C. $2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4$

D. $N_2O_5 + H_2O \rightarrow 2HNO_3$

Answer: A

Solution:

Solution:
We are aware that normally rain water has a pH of 5.6 due to the presence of H^+ions formed by the reactions of rain water with carbon dioxide present in the atmosphere.
$H_2O(l) + CO_2(g) \leftrightharpoons H_2CO_3(aq)$
$H_2CO_3(aq) \leftrightharpoons H^+(aq) + HCO_3^-(aq)$

Question8

How can photochemical smog be controlled?
[1-Feb-2023 Shift 1]

Options:

A. By using tall chimneys

B. By complete combustion of fuel

C. By using catalytic converters in the automobiles/industry

D. By using catalyst

Answer: C

Solution:

Solution:
NCERT (Environmental chemistry)

Question9

The industrial activity held least responsible for global warming is : [1-Feb-2023 Shift 2]

Options:

A. manufacturing of cement

B. steel manufacturing

C. Electricity generation in thermal power plants.

D. Industrial production of urea

Answer: D

Solution:

Solution:
In urea production NH_3 and CO_2 consumed so least responsible for global warming.

Question10

The possibility of photochemical smog formation is more at [6-Apr-2023 shift 1]

Options:

A. The places with healthy vegetation

B. Himalayan villages in winter

C. Marshy lands

D. Industrial areas

Answer: D

Solution:

Solution:
Photochemical smog occurs in warm, dry and sunny climate. The main components come from the action of sunlight on unsaturated hydrocarbon and nitrogen oxides produced by automobiles and factories.

Question11

The group of chemicals used as pesticide is [6-Apr-2023 shift 2]

Options:

A. Sodium chlorate, DDT, PAN

B. Aldrin, Sodium chlorate, Sodium arsinite

C. DDT, Aldrin

D. Dieldrin, Sodium arsinite, Tetrachloroethene

Answer: C

Solution:

Solution:
Pesticides → D.D.T and Aldrin

Question12

Match list I with list II:

List I (species)	List II (Maximum allowed concentration in ppm in drinking water)
A. F^-	I. <50 ppm
B. SO_4^{2-}	II. <5 ppm
C. NO_3^-	III. <2 ppm
D. Zn	IV. <500 ppm

[8-Apr-2023 shift 1]

Options:

A. A-III, B-II, C-I, D-IV

B. A-II, B-I, C-III, D-IV

C. A-IV, B-III, C-II, D-I

D. A-I, B-II, C-III, D-IV

Answer: A

Solution:

Data based

	(Maximum allowed ppm
F^-	<2 ppm
SO_4^{2-}	<5 ppm
NO_3^-	<50 ppm
Zn	<500 ppm

Question13

Which of these reactions is not a part of breakdown of ozone in stratosphere?
[8-Apr-2023 shift 2]

Options:

A. $CF_2Cl_2(g) \xrightarrow{uv} \dot{C}l(g) + \dot{C}F_2Cl(g)$

B. $\dot{C}(g) + O_3(g) \rightarrow Cl\dot{O}(g) + O_2(g)$

C. $2\dot{C}lO \rightarrow ClO_2(g) + \dot{C}l(g)$

D. $\dot{C}lO(g) + O(g) \rightarrow \dot{C}l(g) + O_2(g)$

Answer: C

Solution:

Ozone destruction

$CF_2Cl_2 \xrightarrow{hv} \dot{C}l + \dot{C}F_2Cl(g)$

$\dot{C}l + O_3 \rightarrow Cl\dot{O} + O_2$

$\dot{C}lO + \dot{O} \rightarrow \dot{C}l + O_2$

Question14

Match list I with list II

List I Industry	List II Waste Generated
(A) Steel plants	(I) Gypsum
(B) Thermal power plants	(II) Fly ash
(C) Fertilizer industries	(III) Slag
(D) Paper mills	(IV) Bio-degradable wastes

Choose the correct answer from the options given below
[10-Apr-2023 shift 1]

Options:

A. (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

B. (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

C. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

D. (A) -(III), (B) -(II), (C) -(I), (D) -(IV)

Answer: D

Solution:

Steel plant produces slag from blast furnace. Thermal power plant produces fly ash, Fertilizer industries produces gypsum. Paper mills produces bio degradable waste

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Question15

The delicate balance of CO_2 and O_2 is NOT disturbed by [10-Apr-2023 shift 2]

Options:

A. Burning of Coal

B. Deforestation

C. Burning of petroleum

D. Respiration

Answer: D

Solution:

Solution:
The balance of carbon dioxide and oxygen in atmosphere is mainly maintained by the oxygen released and carbon dioxide consumed during photosynthesis by plants.

Question16

Given below are two statements:
Statement I : If BOD is 4 ppm and dissolved oxygen is 8 ppm, it is a good quality water.
Statement II : If the concentration of zinc and nitrate salts are 5 ppm each, than it can be good quality water.
In the light of the above statements choose the most appropriate answer from the options given below:
[11-Apr-2023 shift 1]

Options:

A. Statement I is incorrect but statement II is correct

B. Statement I is correct but statement II is incorrect

C. Both the statements I and II are incorrect

D. Both the statement I and II are correct

Answer: D

Solution:

Clean water would have BOD value of less than 5 ppm.
Maximum limit of Zn in clean water = 5.0 ppm or $mg\,dm^{-3}$
Maximum limit of NO_3^- in clean water = 50 ppm or $mg\,dm^{-3}$

Question17

Which of the following compounds is an example of Freon? [11-Apr-2023 shift 2]

Options:

A. C_2F_4

B. C_2HF_3

C. $C_2Cl_2F_2$

D. $C_2H_2F_2$

Answer: C

Solution:

Solution:
Freons are chlorofluoro carbon.

Question18

Match list I with list II

LIST I	LIST II
A. Nitrogen oxides in air	I. Eutrophication
B. Methane in air	II. pH of rain water becomes 5.6
C. Carbon dioxide	III. Global warming
D. phosphate fertilisers in water	IV. Acid rain

Choose the correct answer from the options given below:
[12-Apr-2023 shift 1]

Options:

A. A-IV, B-III, C-II, D-I

B. A-I, B-II, C-III, D-IV

C. A-II, B-III, C-I, D-IV

D. A-IV, B-II, C-III, D-I

Answer: A

Solution:

A-IV, B-III, C-II, D-I

Question19

The radical which mainly causes ozone depletion in the presence of UV radiations is :
[13-Apr-2023 shift 1]

Options:

A. NO°

B. $\dot{O}_H$

C. $CH_3^{\cdot}$

D. Cl°

Answer: D

Solution:

$$O_2(g) \xrightarrow{uv} O(g) + O(g)$$
$$O_2(g) + O(g) \rightarrow O_3(g)$$
$$CF_2Cl_2(g) \xrightarrow{uv} \dot{C}l(g) + \dot{C}_2Cl(g)$$
$$\dot{C}l(g) + O_3(g) \rightarrow ClO(g) + O_2(g)$$
$$\dot{C}lO(g) + O(g) \rightarrow \dot{C}l(g) + O_2(g)$$

Question20

Which of the following are the Green house gases?
(A) Water vapour

(B) Ozone
(C) I_2
(D) Molecular hydrogen
Choose the most appropriate answer from the options given below : [13-Apr-2023 shift 2]

Options:

A. C and D only

B. A and B only

C. B and C only

D. A and D only

Answer: B

Solution:

Solution:
Green house gases are CO_2, CH_4, water vapour, nitrous oxide, CFC_s and ozone.

Question21

The possibility of photochemical smog formation will be minimum at [15-Apr-2023 shift 1]

Options:

A. Srinagar, Jammu and Kashmir in January

B. New-Delhi in August (Summer)

C. Kolkata in October

D. Mumbai in May

Answer: A

Solution:

Solution:
Srinage, Jammu & Kashmir in January photo chemical smog is produced when sunlight reacts with nitrogen oxides.

$$NO_2(g) \xrightarrow{hu} NO(g) + O(g)$$
$$O(g) + O_2 \rightarrow O_3(g)$$
$$NO + O_3 \rightarrow NO_2 + O_2$$

Question22

Some gases are responsible for heating of atmosphere (green house

effect). Identify from the following the gaseous species which does not cause it.
[24-Jun-2022-Shift-2]

Options:

A. CH_4

B. O_3

C. H_2O

D. N_2

Answer: D

Solution:

Solution:
Among the given gases, the green house gases which are responsible for heating the atmosphere are CH_4, water vapour and ozone. Nitrogen is not a green house gas.
CH_4, O_3 and H_2O causes global warming in Tropospheric level.
N_2 does not cause global warming.

Question23

The eutrophication of water body results in:
[25-Jun-2022-Shift-1]

Options:

A. loss of Biodiversity.

B. breakdown of organic matter.

C. increase in biodiversity.

D. decrease in BOD.

Answer: A

Solution:

Solution:
Eutrophication is the process in which nutrient enriched water bodies support a dense plant population, which kills animal life by depriving it of oxygen and results in subsequent loss of biodiversity.

Question24

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Polluted water may have a value of BOD of the order of

17 ppm.
Reason R : BOD is a measure of oxygen required to oxidise both the biodegradable an nonbiodegradable organic material in water.
In the light of the above statements, choose the most appropriate answer from the options given below.
[25-Jun-2022-Shift-2]

Options:

A. Both A and R are correct and R is the correct explanation of A.

B. Both A and R are correct but R is NOT the correct explanation of A.

C. A is correct but R is not correct.

D. A is not correct but R is correct.

Answer: C

Solution:

Solution:
Highly polluted water could have a BOD value of 17 ppm or more.
The amount of oxygen required by bacteria to break down the organic matter present in a certain volume of a sample of water is called Biochemical Oxygen demand (BOD).
Hence A is correct but R is not correct.

Question25

Given below are two Statements:
Statement I: Classical smog occurs in cool humid climate. It is a reducing mixture of smoke, fog and sulphur dioxide.
Statement II : Photochemical smog has components, ozone, nitric oxide, acrolein, formaldehyde, PAN etc.
In the light of the above statements, choose the most appropriate answer from the options given below.
[27-Jun-2022-Shift-1]

Options:

A. Both Statement I and Statement II are correct.

B. Both Statement I and Statement II are incorrect.

C. Statement I is correct but Statement II is incorrect.

D. Statement I is incorrect but Statement II is correct.

Answer: A

Solution:

(I) Classical smog occurs in cool humid climate. It is a reducing mixture of smoke, fog and sulphur dioxide. This is a correct statement.

(II) This statement is also based on fact and is a correct statement.

Question26

Which amongst the following is not a pesticide? [28-Jun-2022-Shift-1]

Options:

A. DDT

B. Organophosphates

C. Dieldrin

D. Sodium arsenite

Answer: D

Solution:

Solution:
$NaAsO_2$ (Sodium arsenite) is an insecticide, antibacterial agent herbicide.

Question27

Correct statement about photo-chemical smog is : [28-Jun-2022-Shift-2]

Options:

A. It occurs in humid climate.

B. It is a mixture of smoke, fog and SO_2.

C. It is reducing smog.

D. It results from reaction of unsaturated hydrocarbons.

Answer: D

Solution:

Solution:
Photochemical smog occurs in warm, dry and sunny climate. The main components of photochemical smog result from the action of unsaturated hydrocarbons and nitrogen oxides. This is an oxidising smog.

Question28

Sulphur dioxide is one of the components of polluted air. SO_2 is also a

major contributor to acid rain. The correct and complete reaction to represent acid rain caused by SO_2 is
[29-Jun-2022-Shift-2]

Options:

A. $2SO_2 + O_2 \rightarrow 2SO_3$

B. $SO_2 + O_3 \rightarrow SO_3 + O_2$

C. $SO_2 + H_2O_2 \rightarrow H_2SO_4$

D. $2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4$

Answer: D

Solution:

Solution:
$2SO_2 + O_2 + 2H_2O \longrightarrow 2H_2SO_4$
Acid rain occurs due to increased concentration of oxides of sulphur and Nitrogen.

Question29

The mixture of chloroxylenol and terpineol is an example of
[29-Jun-2022-Shift-2]

Options:

A. antiseptic

B. pesticide

C. disinfectant

D. narcotic analgesic

Answer: A

Solution:

Solution:
The mixture of chloroxylenol and terpineol is known as Dettol. It acts as an Antiseptic.

Question30

The photochemical smog does not generally contain:
[25-Jul-2022-Shift-1]

Options:

A. NO

B. NO_2

C. SO_2

D. $HCHO$

Answer: C

Solution:

Photochemical smog contain:
Ozone, nitric oxide, organic compounds, nitrogen dioxide, formaldehyde.
$\therefore SO_2$ is not the part of photochemical smog.

Question31

Match List I with List II.

List I Pollutant	List II Source
(A) Microorganisms	(I)Strip mining
(B) Plant nutrients	(II)Domestic sewage
(C) Toxic heavy metals	(III) Chemical fertilizer
(D) Sediment	(IV) Chemical factory

Choose the correct answer from the options given below : [26-Jul-2022-Shift-2]

Options:

A. A-II, B-III, C-IV, D-I

B. A-II, B-I, C-IV, D-III

C. A-I, B-IV, C-II, D-III

D. A-I, B-IV, C-III, D-II

Answer: A

Solution:

Pollutant → Source
Microorganisms → Domestic sewage
Plant nutrients → Chemical fertilizers
Toxic heavy metals → Chemical factory
Sediment → Strip mining

Question32

Match List I with List II.

List I Pollutant	List II Disease/sickness
(A) Sulphate (>500 ppm)	(I)Methemoglobinemia
(B) Nitrate (>50 ppm)	(II) Brown mottling of teeth
(C) Lead (>50 ppb)	(III) Laxative effect
(D) Fluoride (>2 ppm)	(IV) Kidney damage

Choose the correct answer from the options given below:
[27-Jul-2022-Shift-1]

Options:

A. A-IV, B-I, C-II, D-III

B. A-III, B-I, C-IV, D-II

C. A-II, B-IV, C-I, D-III

D. A-II, B-IV, C-III, D-I

Answer: B

Solution:

A. Sulphate ($>500\,\text{ppm}$) - Causes Laxative effect that leads to dehydration
B. Nitrate (>50 ppm) - Causes Methemoglobinemia, skin appears blue
C. Lead ($>50\,\text{ppb}$) - It damage kidney and RBC
D. Fluoride ($>2\,\text{ppm}$) - It Causes Brown mottling of teeth

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Question33

Given below are two statements:
Statement I: The non bio-degradable fly ash and slag from steel industry can be used by cement industry.
Statement II : The fuel obtained from plastic waste is lead free.
In the light of the above statements, choose the most appropriate answer from the options given below:
[27-Jul-2022-Shift-2]

Options:

A. Both Statement I and Statement II are correct

B. Both Statement I and Statement II are incorrect

C. Statement I is correct but Statement II is incorrect

D. Statement I is incorrect but Statement II is correct

Answer: A

Solution:

Solution:
Both Statement are correct.
- Fuel obtained from plastic waste has high octane rating. It contain no lead and is known as "green fuel".
- The non bio-degradable fly ash and slag from steel industry can be used by cement industry.

Question34

Given below are two statements:
Statement I : In polluted water values of both dissolved oxygen and BOD are very low.
Statement II : Eutrophication results in decrease in the amount of dissolved oxygen.
In the light of the above statements, choose the most appropriate answer from the options given below:
[28-Jul-2022-Shift-1]

Options:

A. Both Statement I and Statement II are true

B. Both Statement I and Statement II are false

C. Statement I is true but Statement II is false

D. Statement I is false but Statement II is true

Answer: D

Solution:

Solution:
Since eutrophication is result of excessive growth of weed in water bodies, which consume dissolved oxygen of water bodies.
∴ Eutrophication decreases amount of dissolved oxygen in water bodies.
Polluted water has low value of dissolved oxygen, but high valueof BOD (Biological oxygen demand), since chemical and organic matter requires dissolved oxygen to get decompose.

Question35

Which among the following pairs has only herbicides?
[29-Jul-2022-Shift-1]

Options:

A. Aldrin and Dieldrin

B. Sodium chlorate and Aldrin

C. Sodium arsinate and Dieldrin

D. Sodium chlorate and sodium arsinite

Answer: D

Solution:

Both sodium chlorate and sodium arsenate behave as herbicide.

Question36

Which of the following compounds is an example of hypnotic drug? [29-Jul-2022-Shift-1]

Options:

A. Seldane

B. Amytal

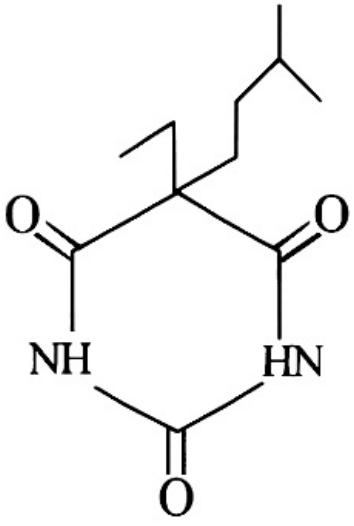
C. Aspartame

D. Prontosil

Answer: B

Solution:

Solution:
Amytal is hypnotic drug used to treat sleeping disorder.



Amytal

Question37

Given below are two statements.
Statement I The pH of rain water is normally ~5.6.
Statement II If the pH of rain water drops below 5.6, it is called acid rain.
In the light of the above statements, choose the correct answer from the

options given below.
[25 Feb 2021 Shift 2]

Options:

A. Both statements I and II are true.

B. Both statements I and II are false.

C. Statement I is true but statement II is false.

D. Statement I is false but statement II is true.

Answer: A

Solution:

Solution:
The pH of rain water is normally 5.6. If it pH drop below 5.6, it is called acid rain.
∴ Both statements are correct.

Question38

Given below are two statements:
Statement I An allotrope of oxygen is an important intermediate in the formation of reducing smog.
Statement II Gases such as oxides of nitrogen and sulphur present in troposphere contribute to the formation of photochemical smog.
In the light of the above statements, choose the correct answer from the options given below.
[25 Feb 2021 Shift 1]

Options:

A. Both statement I and statement II are true.

B. Both statement I and statement II are false.

C. Statement I is true but statement II is false.

D. Statement I is false but statement II is true.

Answer: D

Solution:

Solution:
Reducing smog is a mixture of smoke, fog and sulphur dioxide. Therefore, statement I is false.
Tropospheric pollutants such as nitrogen oxide and sulphur oxide contribute to the formation of photochemical smog. So, statement II is true.

Question39

Given below are two statements.
Statement I The value of the parameter "Biochemical Oxygen Demand (BOD)" is important for survival of aquatic life.
Statement II The optimum value of BOD is 6.5ppm.
In the light of the above statements, choose the most appropriate answer from the options given below.
[24 Feb 2021 Shift 2]

Options:

A. Statement I is false but statement II is true.

B. Both statements are true.

C. Statement I is true but statement II is false.

D. Both statements are false.

Answer: C

Solution:

Solution:
Statement I is true but statement II is false.
Clean water would have BOD value of less than 5ppm whereas highly polluted water could have a BOD value of 17 ppm or more.
Hence, the value of parameter ' BOD ' is important for survival of aquatic life but optimum value of BOD is 17ppm or more. So, statement II is incorrect.

Question40

The hybridisation and magnetic nature of $[Mn(CN)_6]^{4-}$ and $[Fe(CN)_6]^{3-}$, respectively are
[25 Feb 2021 Shift 1]

Options:

A. d^2sp^3 and paramagnetic

B. sp^3d^2 and diamagnetic

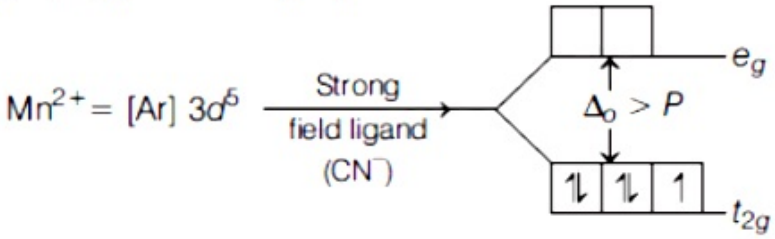
C. d^2sp^3 and diamagnetic

D. sp^3d^2 and paramagnetic

Answer: A

Solution:

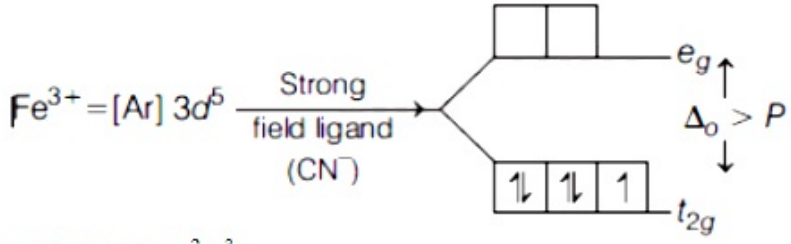
$[Mn(CN)_6]^{4-}$ In this complex, oxidation state of Mn is +2



Hybridisation- d^2sp^3

Magnetic nature-Paramagnetic

$[Fe(CN)_6]^{-3}$



Hybridisation- d^2sp^3

Magnetic nature-Paramagnetic

Question41

The calculated magnetic moments (spin only value) for species $[FeCl_4]^{2-}$, $[Co(C_2O_4)_3]^{3-}$ and MnO_4^{2-} respectively are [24 Feb 2021 Shift 2]

Options:

A. 5.82, 0 and 0BM

B. 4.90, 0 and 1.73BM

C. 5.92, 4.90 and 0BM

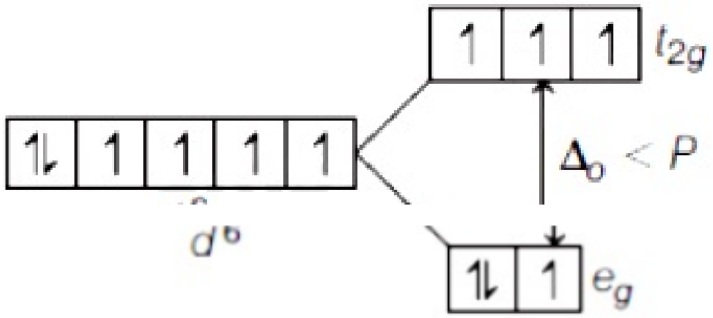
D. 4.90, 0 and 2.83BM

Answer: B

Solution:

(i) $[FeCl_4]^{2-} \longrightarrow Fe^{2+} \longrightarrow [Ar]3d^6$

$\therefore Cl$ is weak field ligand so does not pairing occur



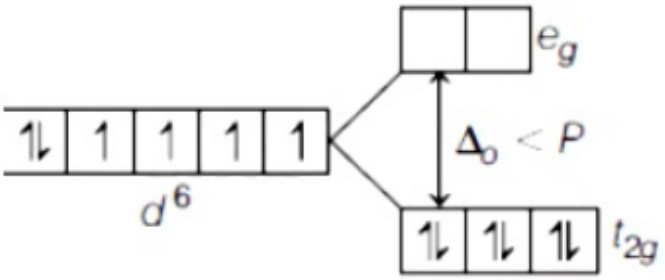
So, magnetic moment $(\mu) = \sqrt{n(n+2)}$ BM $= \sqrt{4(4+2)}$ BM

($n =$ Number of total unpaired $e^- = 4$)

$= \sqrt{24}BM = 4.90BM$

(ii) $[Co(C_2O_4)_3]^{3-} \longrightarrow Co^{3+} \longrightarrow [Ar]3d^6$

C_2O_4 is strong field ligand so pairing occur.



All electrons are paired, $n = 0$

hence, $\mu = 0$

(iii) $MnO_4^{2-} \longrightarrow Mn^{+6} \longrightarrow [Ar]Bd^1$

$n = 1$

$= \sqrt{n(n+2)}BM$

$= \sqrt{1(1+2)}BM$

$= \sqrt{3}BM$

$= 1.73BM$

Question42

The gas released during anaerobic degradation of vegetation may lead to:
[24 Feb 2021 Shift 1]

Options:

A. Ozone hole

B. Acid rain

C. Corrosion of metals

D. Global warming and cancer

Answer: D

Solution:

Solution:
The gas, CH_4 evolved due to anaerobic degradation of vegetation which causes global warming and cancer.

Question43

The statements that are true.
(A) Methane leads to both global warming and photochemical smog.
(B) Methane is generated from paddy fields (C) Methane is a stronger global warming gas than CO_2.
(D) Methane is a part of reducing smog.
Choose the most appropriate answer from the options given below
[18 Mar 2021 Shift 1]

Options:

A. (A), (B), (C) only

B. (A) and (B) only

C. (B), (C), (D) only

D. (A), (B), (D) only

Answer: A

Solution:

Solution:
Methane leads to both global warming and photochemical smog. Order of contribution of global warming gas
Order of contribution of global warming gas
$CO_2 > CH_4 > CFC > O_3 > N_2O > H_2O$

- Methane is generated in large amounts from paddy fields. CH_4 is 40 times stronger greenhouse gas than CO_2.
- CO_2 can be absorbed by photosynthesis or by formation of acid rain etc., while no such activities are there for methane.
- It has more heating effects. Hence, methane is stronger global warming gas than CO_2.
- Methane is not a part of reducing smog.

Hence, statements (A), (B) and (C) are true.

Question44

Reducing smog is a mixture of
[17 Mar 2021 Shift 1]

Options:

A. smoke, fog and O_3

B. smoke, fog and SO_2

C. smoke, fog and $CH_2 = CH - CHO$

D. smoke, fog and N_2O_3

Answer: B

Solution:

Solution:
Classical smog (reducing smog) occurs in cool humid climate, it is a mixture of smoke, fog and SO_2.
It is a reducing mixture/smog.
Reducing smog = smoke + fog + SO_2
Note Photochemical smog occurs in warm, dry and sunny climate. The main components of the photochemical smog result from the action of sunlight on unsaturated hydrocarbons and nitrogen oxides produced by automobiles and factories. Photochemical smog has high concentration of oxidising agents and is therefore, called as oxidising smog.

Question45

The green house gas/es is (are)
(A) carbon dioxide
(B) oxygen
(C) water vapour
(D) methane
Choose the most appropriate answer from the options given below.
[16 Mar 2021 Shift 2]

Options:

A. Only (A) and (C)

B. Only (A)

C. (A), (C) and (D)

D. Only (A) and (B)

Answer: C

Solution:

Solution:
Greenhouse gases are carbon dioxide, methane, water vapours, nitrous oxide, CFCs and ozone. The greenhouse effect is a natural process that warms the Earth's surface. When the Sun's energy reaches the Earth's atmosphere, some of it is reflected back to space and the rest is absorbed and re-radiated by greenhouse gases. The absorbed energy warms the atmosphere and the surface of the Earth.

Question46

The type of pollution that gets increased during the day time and in the presence of O_3 is
[16 Mar 2021 Shift 1]

Options:

A. reducing smog

B. oxidising smog

C. global warming

D. acid rain

Answer: B

Solution:

Solution:
Oxidising smog (photochemical smog) gets increased during the day time and in the presence of O_3.
It occurs in warm, dry and sunny climate. When fossil fuels are burnt, hydrocarbons (unburnt fuels) and nitric oxide (NO) are produced. Ozone reacts rapidly with the NO to generate NO_2.
$NO(g) + O_3(g) \longrightarrow NO_2(g) + O_2(g)$
Ozone is a toxic gas and both NO_2 and O_3 are strong oxidising agents and can react with the unburnt hydrocarbons in the polluted air to produce chemicals such as formaldehyde, acrolein and peroxyacetyl nitrate (PAN).

Question47

Given below are two statements
Statement I Non-biodegradable wastes are generated by the thermal power plants.
Statement II Bio-degradable detergents leads to eutrophication.
In the light of the above statements,
choose the most appropriate answer from the option given below.
[18 Mar 2021 Shift 2]

Options:

A. Both statements I and II are false.

B. Statement 1 is true but statement I is false.

C. Statement I is false but statement II is true.

D. d. Both statements I and II are true.

Answer: D

Solution:

Solution:
Non-biodegradable wastes are generated by the thermal power plants which produces fly ash. Detergents which are biodegradable causes problem called eutrophication, which kills aquatic life by deprieving it of oxygen.
Hence, both statements are true.

Question48

Which of the following statement(s) is (are) incorrect reason for eutrophication?
A. Excess usage of fertilisers.
B. Excess usage of detergents.
C. Dense plant population in water bodies.
D. Lack of nutrients in water bodies that prevent plant growth.
Choose the most appropriate answer from the options given below.
[17 Mar 2021 Shift 2]

Options:

A. Only A

B. Only C

C. B and D

D. Only D

Answer: D

Solution:

Solution:
Statement (D) is incorrect.
As eutrophication is the gradual increase in the concentration of phosphorus, nitrogen, and other plant nutrients in an aging aquatic ecosystem such as a lake. The productivity or fertility of such an ecosystem naturally increases as the amount of organic material that can be broken down into nutrients increases.

Question49

Arrange the following metal complex\/compounds in the increasing order of spin only magnetic moment. Presume all the three, high spin system.
(Atomic numbers Ce = 58, Gd = 64 and Eu = 63.)

A. $(NH_4)_2[Ce(NO_3)_6]$
B. $Gd(NO_3)_3$ and
C. $Eu(NO_3)_3$
[16 Mar 2021 Shift 2]

Options:

A. (B) < (A) < (C)

B. (C) < (A) < (B)

C. (A) < (B) < (C)

D. (A) < (C) < (B)

Answer: D

Solution:

(A) $_{58}Ce \longrightarrow [Xe]4f^{2}5d^{0}6s^{2}$
In complex $Ce^{+4} \longrightarrow [Xe]4f^{0}5d^{0}6s^{0}$
There is no unpaired electron, so $\mu_m = 0BM$
(B) $_{64}Gd \longrightarrow [Xe]4f^{7}5d^{1}6s^{2}$
In complex, $_{64}Gd^{2+} \longrightarrow [Xe]4f^{7}5d^{1}$
There are 8 unpaired electrons.
$\therefore \ \mu_m = \sqrt{8(8+2)} = \sqrt{80} = 8.94BM$
(C) $_{63}Eu \longrightarrow [Xe]4f^{9}5d^{0}6s^{0}$
In complex $_{63}Eu^{+3} \longrightarrow [Xe]4f^{6}5d^{0}6s^{0}$
contain six unpaired electrons.
So, $\mu_m = \sqrt{6(6+2)} = \sqrt{48}BM = 6.93BM$
Hence, order of spin only magnetic moment B > C > A

Question50

Match List - I with List - II :

	List - I		List - II
(a)	Carbon monoxide	(i)	Carcinogenic
(b)	Sulphur dioxide	(ii)	Metabolized bypyrus plants
(c)	Polychlorinated biphenyls	(iii)	Haemoglobin
(d)	Oxides of Nitrogen	(iv)	Stiffness offlower buds

Choose the correct answer from the options given below :
[27 Jul 2021 Shift 2]

Options:

A. (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

B. (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

C. (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

D. (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

Answer: A

Solution:

Question51

Given below are two statements :
Statement I : Chlorofluoro carbons breakdown by radiation in the visible energy region and release chlorine gas in the atmosphere which then reacts with stratospheric ozone.
Statement II : Atmospheric ozone reacts with nitric oxide to give nitrogen and oxygen gases, which add to the atmosphere.
For the above statements choose the correct answer from the options given below :
[25 Jul 2021 Shift 2]

Options:

A. Statement I is incorrect but statement II is true

B. Both statement I and II are false

C. Statement I is correct but statement II is false

D. Both statement I and II are correct

Answer: B

Solution:

Statement (1)
CFCs are broken down by powerful UV radiation and releases chlorine free radical which reacts with ozone and start chain reaction.

$$CF_2Cl_{2(g)} \xrightarrow{UV} \dot{Cl}_{(g)} + \dot{C}F_2Cl_{(g)}$$

$$\dot{Cl}_{(g)} + O_{3(g)} \rightarrow \dot{Cl}O_{(g)} + O_{2(g)}$$

$$\dot{C}O_{(g)} + O_{(g)} \rightarrow \dot{Cl}_{(g)} + O_{2(g)}$$

Statement (2)
Atmosphere ozone reacts with nitric oxide to produce nitrogen dioxide and oxygen.

$$NO_{(g)} + O_{3(g)} \rightarrow NO_{2(g)} + O_{2(g)}$$

Question52

Which one of the following gases is reported to retard photosynthesis?
[20 Jul 2021 Shift 2]

Options:

A. CO

B. CF Cs

C. CO_2

D. NO_2

Answer: D

Solution:

Solution:
According to NCERT only NO_2 from the given options can retard the photosynthesis process in plants.

Question53

Which one of the following statements is NOT correct ? [27 Jul 2021 Shift 1]

Options:

A. Eutrophication indicates that water body is polluted ?

B. The dissolved oxygen concentration below 6 ppm inhibits fish growth

C. Eutrophication leads to increase in the oxygen level in water

D. Eutrophication leads to anaerobic conditions

Answer: C

Solution:

Solution:
Eutrophication leads to decrease in oxygen level of water.
3rd statement is incorrect

Question54

In stratosphere most of the ozone formation is assisted by [27 Aug 2021 Shift 2]

Options:

A. cosmic rays

B. γ-rays

C. ultraviolet radiations

D. visible radiations

Answer: C

Solution:

In the stratosphere, the UV-light causes the splitting of O_2 molecule into oxygen atoms. These atoms react with oxygen present in the air to form ozone.

$$O_2(g) \xrightarrow{\text{UV radiation}} O(g) \xrightarrow[\text{Ozone}]{O_2(g) + \text{UV radiation}} O_3(g)$$

Therefore, the option (c) is correct.

Question55

The gas 'A' is having very low reactivity reaches to stratosphere. It is non-toxic and non-flammable but dissociated by UV-radiations in stratosphere. The intermediates formed initially from the gas 'A' are [27 Aug 2021 Shift 1]

Options:

A. $ClO + CF_2Cl$

B. $ClO + CH_3$

C. $CH_3 + CF_2Cl$

D. $Cl + CF_2Cl$

Answer: D

Solution:

The gas is chlorofluoro carbon. Its reaction in stratosphere is as follows :

(i) $CCl_2F_2 + \text{UV light} \rightarrow CClF_2 + Cl$

(ii) $CCl_3F + \text{UV light} \rightarrow CCl_2F + Cl$

(iii) $Cl + O_3 \rightarrow ClO + O_2$

(iv) $ClO + O \rightarrow Cl + O_2$

So, the intermediate in first step are $Cl + CClF_2$.

Question56

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) Photochemical smog causes cracking of rubber.
Reason (R) Presence of ozone, nitric oxide, acrolein, formaldehyde and peroxyacetyl nitrate in photochemical smog makes it oxidising.
Choose the most appropriate answer from the options given below. [26 Aug 2021 Shift 2]

Options:

A. Both (A) and (R) are true but (R) is not the correct explanation of (A).

B. Both (A) and (R) are true and (R) is the correct explanation of (A).

C. (A) is false but (R) is true.

D. (A) is true but (R) is false.

Answer: B

Solution:

Solution:
Both Assertion and Reason are true and Reason is the correct explanation of Assertion. Smog is a type of intense air pollution. It is composed of nitrogen oxides, sulphur oxides, ozone, smoke and other particulates. Rubber is an organic compound and it is sensitive to number of chemical influences-oxidation, reduction etc. Excessive cracking of rubber products can be one of the indicators of the presence of atmospheric photochemical oxidants. Excessive cracking of rubber is caused by atmospheric ozone formed in the photochemical smog formation process.

Question57

BOD values (in ppm) for clean water (A) and polluted water (B) are expected respectively
[31 Aug 2021 Shift 1]

Options:

A. A > 50, B < 27

B. A > 25, B < 17

C. A < 5, B > 17

D. A > 15, B > 47

Answer: C

Solution:

BOD value of clean water is less than 5 ppm.
So, A < 5
BOD value of polluted water is greater than 17 ppm.
So, B > 17
Thus, correct option is (c).

Question58

Water sample is called cleanest on the basis of which one of the BOD values given below
[1 Sep 2021 Shift 2]

Options:

A. 11 ppm

B. 15 ppm

C. 3 ppm

D. 21 ppm

Answer: C

Solution:

Solution:
BOD is the biological oxygen demand. Cleanest water sample will have BOD value equal to (SO_2)
3 ppm as clean water could have BOD value of less than 5 ppm.

Question59

Among the gases, (a)-(e) the gases that cause greenhouse effect are:
(1) CO_2
(2) H_2O
(3) CF Cs
(4) O_2
(5) O_3
[Jan. 08, 2020 (I)]

Options:

A. (1), (2), (3) and (4)

B. (1), (2), (3) and (5)

C. (1) and (4)

D. (1), (3), (4) and (5)

Answer: B

Solution:

Solution:
CO_2, O_3, H_2O vapours and CFC's are the gases that cause green house effect.

Question60

Biochemical Oxygen Demand (BOD) is the amount of oxygen required (in ppm):
[Jan. 09, 2020 (II)]

Options:

A. for sustaining life in a water body.

B. by bacteria to break-down organic waste in a certain volume of a water sample.

C. for the photochemical breakdown of waste present in 1m^3 volume of a water body.

D. by anaerobic bacteria to breakdown inorganic waste present in a water body.

Answer: B

Solution:

Biochemical oxygen demand is the amount of oxygen required by bacteria to break down the organic matter present in a given volume of a sample of water.

Question61

The processes of calcination and roasting in metallurgical industries, respectively, can lead to :
[Sep. 04, 2020 (II)]

Options:

A. Global warming and photochemical smog

B. Global warming and acid rain

C. Photochemical smog and ozone layer depletion

D. Photochemical smog and global warming

Answer: B

Solution:

Solution:
Calcination : Heating of concentrated ore in the absence (or limited supply of air). Mainly used for carbonate ores.

$$\text{Metal carbonate} \xrightarrow{\Delta} \text{Metal oxide} + \underset{\text{(causes global warming)}}{CO_2}$$

Roasting : Heating of concentrated ore in a regular supplyof air. Mainly used for sulphide ores.

$$\text{Metal sulphide} + O_2 \longrightarrow \text{Metal oxide} + \underset{\text{(causes acid rain)}}{SO_2}$$

Question62

Thermal power plants can lead to:
[Sep. 03, 2020(I)]

Options:

A. Acid rain

B. Blue baby syndrome

C. Ozone layer depletion

D. Eutrophication

Answer: A

Solution:

Solution:
Burning of fossil fuels (which contain sulphur and nitrogenous matter) such as coal and oil in power stations and furnaces produce sulphur dioxide and nitrogen oxides which causes acid rain.

Question63

The incorrect statement(s) among (1) − (4) regarding acid rain is (are):
(1) It can corrode water pipes.
(2) It can damage structures made up of stone.
(3) It cannot cause respiratory ailments in animals
(4) It is not harmful for trees
[Sep. 03, 2020 (II)]

Options:

A. (1), (3) and (4)

B. (3) only

C. (1), (2) and (4)

D. (3) and (4)

Answer: D

Solution:

Solution:
Acid rain is harmful for trees and animal life. It can also cause respiratory ailments in animals.

Question64

The statement that is not true about ozone is :
[Sep. 02,2020(l)]

Options:

A. in the stratosphere, CFCs release chlorine free radicals (Cl) which reacts with O_3 to give chlorine dioxide radicals.

B. in the atmosphere, it is depleted by CFCs.

C. in the stratosphere, it forms a protective shield against UV radiation.

D. it is a toxic gas and its reaction with NO gives NO_2.

Answer: A

Solution:

In presence of sunlight CFC's molecule divides and release chlorine free radical, which react with ozone give chlorine monoxide radical (ClO^*) and oxygen.

$$CF_2Cl_2(g) \xrightarrow{UV} \dot{C}l(g) + \dot{C}F_2Cl(g)$$

$$Cl^*(g) + O_3(g) \longrightarrow ClO^*(g) + O_2(g)$$

$$ClO^*(g) + O(g) \longrightarrow Cl^*(g) + O_2(g)$$

Question65

The presence of soluble fluoride ion upto 1ppm concentration in drinking water, is:
[Sep. 06,2020 (I)]

Options:

A. harmful for teeth

B. harmful to skin

C. harmful to bones

D. safe for teeth

Answer: D

Solution:

Solution:

Fluoride (F^-) ion conc. upto 1ppm makes the enamel of teeth much harder by converting

$$\underset{\text{hydroxyapatite}}{[3(Ca_3(PO_4)_2)\cdot Ca(OH)_2]} \xrightarrow{F^-} \underset{\text{Fluorapatite(harder)}}{[3(Ca_3(PO_4)_2)\cdot CaF_2]}$$

But above 2 ppm cause brown motting of teeth.

Question66

The condition that indicates a polluted environment is:
[Sep. 05,2020(I)]

Options:

A. eutrophication

B. 0.03% of CO_2 in the atmosphere

C. BOD value of 5ppm

D. pH of rain water to be 5.6

Answer: A

Solution:

Solution:
The process in which nutrient enriched water bodies support a dense plant population which kill animal life by depriving it of oxygen results in subsequent loss of biodiversity is known as Eutrophication. So, this condition indicates a polluted environment.

Question67

The upper stratosphere consisting of the ozone layer protects us from the sun's radiation that falls in the wavelength region [Jan.12, 2019 (II)]

Options:

A. 200 − 315nm

B. 400 − 550nm

C. 0.8 − 1.5nm

D. 600 − 750nm

Answer: A

Solution:

Solution:
Wavelength range lies in 200nm − 315nm (U.V light)

Question68

The molecule that has minimum\/no role in the formation of photochemical smog, is: [Jan. 12, 2019 (I)]

Options:

A. N ,

B. $CH_2 = O$

C. O_3

D. NO

Answer: A

Solution:

NO, O_3 and $HCHO$ are responsible for the formation of photochemical smog while N_2 has no role in photochemical smog.

Question69

The compound that is NOT a common component of photochemical smog is :
[Jan. 12, 2019 (II)]

Options:

A. O_3

B. $H_3C-\underset{\underset{O}{\|}}{C}-OONO_2$

C. $CH_2=CHCHO$

D. CF_2Cl_2

Answer: D

Solution:

Solution:
Acrolein, PAN (produced by the oxidation of unburnt hydrocarbons present in the polluted air) and ozone are the main components of photochemical smog. While CF_2Cl_2 is not a common component of photochemical smog.

Question70

Peroxyacetyl nitrate (PAN), an eye irritant, is produced by:
[Jan. 11, 2019(I)]

Options:

A. Classical smog

B. Acid rain

C. Organic waste

D. Photochemical smog

Answer: D

Solution:

Solution:
P.A.N is produced by photochemical smog.

Question71

Taj Mahal is being slowly disfigured and discoloured. This is primarily due to :
[Jan. 11,2019 (II)]

Options:

A. Global warming

B. Acidrain

C. Water pollution

D. Soil pollution

Answer: B

Solution:

Solution:
Acidrain

Question72

The higher concentration of which gas in air can cause stiffness of flower buds?
[Jan. 11,2019 (II)]

Options:

A. N O_2

B. CO_2

C. SO_2

D. CO

Answer: C

Solution:

Solution:
High concentration of SO_2 leads to stiffness of flow- er buds.

Question73

The reaction that is NOT involved in the ozone layer depletion mechanism in the stratosphere is:
[Jan. 10,2019 (II)]

Options:

A. $CF_2Cl_2(g) \xrightarrow{uv} \dot{C}l(g) + \dot{C}F_2Cl(g)$

B. $ClO(g) + O(g) \longrightarrow \dot{C}1(g) + O_2(g)$

C. $CH_4 + 2O_3 \longrightarrow 3CH_2 = O + 3H_2O$

D. $HOCl(g) \xrightarrow{hv} \dot{O}H(g) + \dot{C}l(g)$

Answer: C

Solution:

Solution:
The reaction, $CH_4 + 2O_3 \rightarrow 3CH_2 = O + 3H_2O$, is not involved in the ozone layer depletion mechanism.

Question74

Water samples with BOD values of 4ppm and 18ppm, respectively, are: [Jan. 12, 2019 (I)]

Options:

A. Clean and Clean

B. Highly polluted and Clean

C. Clean and Highly polluted

D. Highly polluted and Highly polluted

Answer: C

Solution:

Solution:
Clean water would have BOD value $<5ppm$ whereas highly polluted water could have BOD value 17 ppm.

Question75

The concentration of dissolved oxygen (DO) in cold water can go upto : [Jan. 11, 2019 (I)]

Options:

A. 14ppm

B. 8ppm

C. 10ppm

D. 16ppm

Answer: C

Solution:

Dissolved oxygen (DO) in cold water can reach a concentration upto 10ppm.

Question76

Water filled in two glasses A and B have BOD values of 10 and 20 , respectively. The correct statement regarding, them, is: [Jan. 10,2019 (I)]

Options:

A. B is more polluted than A.

B. A is suitable for drinking, whereas B is not.

C. Both A and B are suitable for drinking.

D. A is more polluted than B.

Answer: A

Solution:

Solution:
BOD indicates the amount of putrescible organic matter present in water. Therefore, a low BOD is an indicator of good quality water while a high BOD indicates polluted water.

Question77

A water sample has ppm level concentration of the following metals: F e = 0.2; M n = 5.0; Cu = 3.0; Z n = 5.0. The metal that makes the water sample unsuitable for drinking is: [Jan. 9, 2019 (I)]

Options:

A. Cu

B. M n

C. Fe

D. Z n

Answer: B

Solution:

The water sample containing $Mn = 5ppm$ is unsuitable for drinking as the prescribed level for Mn in drinking water is 0.5ppm.

Question78

Which of the following conditions in drinking water causes methemoglobinemia? [Jan. 9, 2019 (II)]

Options:

A. >50 ppm of lead

B. >50 ppm of chloride

C. >50 ppm of nitrate

D. >100 ppm of sulphate

Answer: C

Solution:

Solution:
Methemoglobinemia is caused by drinking water which is contaminated with nitrate.

Question79

The pH of rain water, is approximately: [Jan. 9, 2019 (II)]

Options:

A. 5.6

B. 7.5

C. 7.0

D. 6.5

Answer: A

Solution:

Solution:
pH of rain water is around 5.6

Question80

The correct set of species responsible for the photochemical smog is: [April 12, 2019 (I)]

Options:

A. N_2, NO_2 and hydrocarbons

B. CO_2, NO_2, SO_2 and hydrocarbons

C. NO, NO_2, O_3 and hydrocarbons

D. N_2, O_2, O_3 and hydrocarbons

Answer: C

Solution:

Solution:
Photochemical smog contains oxides of nitrogen, ozone and hydrocarbons. It is also called oxidation smog.

Question81

The primary pollutant that leads to photochemical smog is: [April 12, 2019 (II)]

Options:

A. acrolein

B. nitrogen oxides

C. ozone

D. sulphur dioxide

Answer: B

Solution:

Solution:
In photochemical smog, NO_2 and hydrocarbons are primary pollutants whereas ozone and acrolein are secondary pollutants.

Question82

The regions of the atmosphere, where clouds are formed and where we live, respectively, are: [April 10, 2019 (I)]

Options:

A. Troposphere and Stratosphere

B. Stratosphere and Troposphere

C. Troposphere and Troposphere

D. Stratosphere and Stratosphere

Answer: C

Solution:

Solution:
The lowest region of atmosphere in which human beings live and clouds form is known as troposphere. It expends up to a height of 10km from sea level.

Question83

Air pollution that occurs in sunlight is : [April 10, 2019 (II)]

Options:

A. reducing smog

B. acid rain

C. oxidizing smog

D. fog

Answer: C

Solution:

Solution:
Air pollution caused by sunlight is photochemical smog also knows as oxidising smog.

Question84

Excessive release of CO_2 into the atmosphere results in: [April 9, 2019 (I)]

Options:

A. global warming

B. polar vortex

C. formation of smog

D. depletion of ozone

Answer: A

Solution:

Global warming is caused by the emission of green house gases. 72% of the totally emitted green house gases is CO_2. Therefore, excessive release of CO_2 is the main cause of global warming. (c) Stratosphere lies between $10 - 50\text{km}$ above sea level.

Question85

The layer of atmosphere between 10km to 50km above the sea level is called as:
[April 9, 2019 (II)]

Options:

A. troposphere

B. thermosphere

C. stratosphere

D. mesosphere

Answer: C

Solution:

Solution:
Stratosphere lies between $10 - 50\text{km}$ above sea level.

Question86

Which is wrong with respect to our responsibility as a human being to protect our environment?
[April 8,2019 (I)]

Options:

A. Restricting the use of vehicles

B. Avoiding the use of floodlighted facilities

C. Setting up compost in gardens.

D. Using plastic bags.

Answer: D

Solution:

Solution:
Use of plastic bags is hazardous to our environment.

Question87

Assertion : Ozone is destroyed by CFCs in the upper stratosphere.
Reason : Ozone holes increase the amount ofUV radiation reaching the earth.
[April 8, 2019 (I)]

Options:

A. Assertion and reason are incorrect.

B. Assertion and reason are both correct, and the reason is the correct explanation for the assertion.

C. Assertion and reason are correct, but the reason is not the explanation for the assertion.

D. Assertion is false, but the reason is correct.

Answer: C

Solution:

CFC's are responsible for depletion of ozone layer

$Cl + O_3 \xrightarrow{uv} \dot{C}l\,(g) + F_2\dot{C}Cl\,(g)$

$Cl + O_3 \longrightarrow Cl\,\dot{O}(g) + O_2(g)$

$Cl\,\dot{O} + O \longrightarrow Cl + O_2(g)$

Due to depletion of ozone layer more UV radiation filters into troposphere.

Question88

The maximum prescribed concentration of copper in drinking water is:
[April 8, 2019 (II)]

Options:

A. 5ppm

B. 0.05ppm

C. 0.5ppm

D. 3ppm

Answer: D

Solution:

Solution:
Maximum prescribed concentration of Cu in drinking water is 3ppm.

Question89

The recommended concentration of fluoride ion in drinking water is up to 1ppm as fluoride ion is required to make teeth enamel harder by converting
$[3Ca_3(PO_4)_2 \cdot Ca(OH)_2]$ to :
[2018]

Options:

A. $[CaF_2]$

B. $[3(CaF_2) \cdot Ca(OH)_2]$

C. $[3Ca_3(PO_4)_2 \cdot CaF_2]$

D. $[3\{(Ca(OH)_2\} \cdot CaF_2]$

Answer: C

Solution:

$$\underset{\text{Hydroxyapatite}}{[3Ca_3(PO_4)_2 \cdot Ca(OH)_2]} + \underset{\text{(drinking water upto 1 ppm)}}{2F^-} \longrightarrow \underset{\text{Fluorapatite(Harder teeth enamel)}}{[3Ca_3(PO_4)_2 \cdot CaF_2]} + 2OH^-$$

Question90

Biochemical oxygen demand(BOD) value can be a measure of water pollution caused by the organic matter. Which of the following statements is correct?
[Online April 15, 2018 (II)]

Options:

A. Polluted water has BOD value higher than 10ppm

B. Aerobic bacteria decreases the BOD value

C. Anaerobic bacteria increases the BOD value

D. Clean water has BOD value higher than 10 ppm

Answer: A

Solution:

Solution:
Clean water has BOD value less than 5 ppm. Polluted water has BOD value higher than 10ppm.

Question91

Identify the pollutant gases largely responsible for the discoloured and lustreless nature of marble of the Taj Mahal.
[(Online April 8, 2017]

Options:

A. O_3 and CO_2

B. CO_2 and NO_2

C. SO_2 and NO_2

D. SO_2 and O_3

Answer: C

Solution:

Solution:
SO_2 and NO_2 are responsible for the discoloured and lustureless nature of marble.

Question92

Which of the following is a set of green house gases?
[Online April 9, 2017]

Options:

A. CH_4, O_3, N_2, SO_2

B. O_3, N_2, CO_2, NO_2

C. O_3, NO_2, SO_2, Cl_2

D. CO_2, CH_4, N_2O, O_3

Answer: D

Solution:

Solution:
Green house gases are those gases that absorbs and emits radiation within the thermal IRrange. The process is the fundamental cause of the green house effect. These are CO_2, CH_4 N_2O and O_3.

Question93

A water sample has ppm level concentration of following anions
F^- = 10; SO_4^{2-} = 100; NO_3^- = 50
the anion/anions that make/makes the water sample unsuitable for drinking is/are:

[2017]

Options:

A. only NO_3^-

B. both SO_4^{2-} and NO_3^-

C. only F^-

D. only SO_4^{2-}

Answer: C

Solution:

Solution:
Concentration of F^- in drinking water above $2ppm$ cause brown mottling of teeth.

Question94

Which one of the following substances used in dry cleaning is a better strategy to control environmental pollution?
[Online April 10,2016]

Options:

A. Sulphur dioxide

B. Carbon dioxide

C. Nitrogen dioxide

D. Tetrachloroethylene

Answer: D

Solution:

Solution:
Tetrachloroethylene is widely used for dry-cleaning fabrics and metal degreasing operations.

Question95

BOD stands for:
[Online April 9, 2016]

Options:

A. Biochemical Oxidation Demand

B. Biological Oxygen Demand

C. Biochemical Oxygen Demand

D. Bacterial Oxidation Demand

Answer: B

Solution:

BOD stands for Biological Oxygen Demand

Question96

The concentration of fluoride, lead, nitrate and iron in a water sample from an underground lake was found to be 1000ppb, 40ppb, 100ppm and 0.2ppm, respectively. This water is unsuitable for drinking due to high concentration of:
[2016]

Options:

A. Nitrate

B. Iron

C. Fluoride

D. Lead

Answer: A

Solution:

Solution:
The maximum limit of nitrate in drinking water is 50 ppm. Excess nitrate in drinking water can cause disease such as methemoglobinemia ('blue baby' syndrome).

Question97

Photochemical smog consists of excessive amount of X , in addition to aldehydes, ketones, peroxyacetyl nitrate (PAN), and so forth X is :
[Online April 10, 2015]

Options:

A. CO

B. CH_4

C. O_3

D. CO_2

Answer: C

Solution:

Chemical pollutants in photochemical smog are nitrogen oxides (NO and NO_2), volatile organic compounds, ozone (O_3) and peroxyacetyl nitrate.

Question98

Addition of phosphate fertilisers to water bodies causes:
[Online April 11,2015]

Options:

A. increase in amount of dissolved oxygen in water

B. deposition of calcium phosphate

C. increase in fish population

D. enhanced growth of algae

Answer: D

Solution:

Solution:

Question99

Which of the following statements about the depletion of ozone layer is correct?
[Online April 11, 2014]

Options:

A. The problem of ozone depletion is less serious at poles because NO_2 solidifies and is not available for consuming ClO^* radicals.

B. The problem of ozone depletion is more serious at poles because ice crystals in the clouds over poles act as catalyst for photochemical reactions involving the decomposition of ozone of Cl^* and ClO^* radicals.

C. Freons, chlorofluorocarbons, are inert. Chemically, they do not react with ozone in stratosphere.

D. Oxides of nitrogen also do not react with ozone in stratosphere.

Answer: B

Solution:

$CF\,Cl_2 + U\,V\ \ light \longrightarrow CF\,Cl_2 + Cl\,.$
$Cl\ + O_3 \longrightarrow Cl\,O + O_2$
$Cl\,O + O\cdot \longrightarrow Cl\ + O_2$
Reaction without polar stratospheric clouds
$CF\,Cl_2 + U\,V\ \ light \rightarrow Cl\cdot + CF\,Cl_2$
$Cl\cdot + O_3 \rightarrow Cl\,O\cdot + O_2$
$Cl\,O\cdot + N\,O_2 \rightarrow Cl\,ON\,O_2$ (Reservoirs)
With polar stratospheric clouds
$$\underset{+HCl}{Cl\,ON\,O_2} + \underset{\text{Moisture}}{\text{Clouds}} \rightarrow \underset{+}{Cl_2} \rightarrow Cl\ \cdot \xrightarrow[+]{O_3} Cl\,O^*$$
The main reason for ozone depletion in polar region is that ice crystals catalyse ozone depletion reaction.

Question100

Global warming is due to increase of:
[Online April 12, 2014]

Options:

A. methane and nitrous oxide in atmosphere

B. methane and CO_2 in atmosphere

C. methane and O_3 in atmosphere

D. methane and CO in atmosphere

Answer: B

Solution:

Solution:
The main gas responsible for the global warming is carbon dioxide. Other contributors include methane released from landfills and agriculture.

Question101

The gas leaked from a storage tank of the Union Carbide plant in Bhopal gas tragedy was:
[2013]

Options:

A. Methyl isocyanate

B. Methylamine

C. Ammonia

D. Phosgene

Answer: A

Question102

Green house gases can be arranged in 'Global Warming Potential' sequence as
[Online May 7, 2012]

Options:

A. $N_2O > CFC > CH_4 > CO_2$

B. $CFC > N_2O > CH_4 > CO_2$

C. $CFC > CO_2 > N_2O > CH_4$

D. $CO_2 > CFC > N_2O > CH_4$

Answer: B

Solution:

Solution:
Global Worming Potential (GWP) is a measure of how much heat a green house gas traps in the atmosphere upto a specific time horizon, relative to CO_2. GWP for CO_2 is standardized to 1 and relative GWP for green house gases are measuredw.r. to CO_2. A high GWP carrecates with a large IR absorption and a long atmospheric life time. Thus, the sequence is
$CFC > N_2O > CH_4 > CO_2$

Question103

Which one of the following depletes ozone layer?
[Online May 26, 2012]

Options:

A. CO

B. NO and freons

C. SO_2

D. CO_2

Answer: B

Solution:

N O and freons are responsible for ozone layer depletion.

Question104

Water sample is reported to be highly polluted if BOD (Biological Oxygen Demand) value of sample becomes [Online May 12, 2012]

Options:

A. more than 17ppm

B. equal to 10ppm

C. equal to 5 ppm

D. less than 5ppm

Answer: A

Solution:

Solution:
Water sample is reported to be highly polluted if BOD value of sample is more than 17 ppm.

Question105

What is DDT among the following? [2012]

Options:

A. Greenhouse gas

B. Afertilizer

C. Biodegradable pollutant

D. Non-biodegradable pollutant

Answer: D

Solution:

Solution:
DDT is a non-biodegradable pollutant.

Question106

Identify the incorrect statement from the following:

[2011RS]

Options:

A. Ozone absorbs the intense ultraviolet radiation of the sun.

B. Depletion of ozone layer is because of its chemical reactions with chlorofluoro alkanes.

C. Ozone absorbs infrared radiation.

D. Oxides of nitrogen in the atmosphere can cause the depletion of ozone layer.

Answer: C

Solution:

Solution:
The ozone layer, existing between 20 to 35km above the earth's surface, shield the earth from the harmful U. V. radiations from the sun.

Question107

Identify the wrong statement in the following:
[2008]

Options:

A. Chlorofluorocarbons are responsible for ozone layer depletion

B. Greenhouse effect is responsible for global warming

C. Ozone layer does not permit infrared radiation from the sun to reach the earth

D. Acid rain is mostly because of oxides of nitrogen and sulphur

Answer: C

Solution:

Solution:
Note: Ozone layer acts as a shield and does not allow ultraviolet radiation from sun to reach earth. It does not prevent infra-red radiation from sun to reach earth.

Question108

The smog is essentially caused by the presence
[2004]

Options:

A. oxides of sulphur and nitrogen

B. O_2 and N_2

C. O_2 and O_3

D. O_3 and N_2

Answer: A

Solution:

Photochemical smog is caused by oxides of sulphur and nitrogen.
